

CytoOrg Guide

Quantifying cytoskeletal organization through skeleton-based
image analysis

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github.com/EP-bioimage-tools/CytoOrg-ImageJ

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1 Introduction

IN ONE SENTENCE CytoOrg converts 2D fluorescence images of filamentous cytoskeletal structures into reproducible measurements of thickness, network geometry, branching, and orientation.

CytoOrg is a single-file Fiji analysis script written in Groovy and run from the Fiji Script Editor. Version 1.0 analyzes one cytoskeletal signal at a time. It can process an isolated cell in each image or several cells in the same field when a matching binary cell mask is provided.

CytoOrg applies the Sato Tubeness filter to the original fluorescence image to enhance tubular structures, then skeletonizes the resulting network into a one-pixel-wide graph of nodes and branches that can be quantified. Because skeletonization removes width information, filament thickness is measured separately before the skeleton is created.

Why skeletonization?

Skeletonization reduces a filament network to its medial axis: a one-pixel-wide centerline that preserves connectivity and overall path geometry. Endpoints and junctions become graph nodes, while the paths between them become segments or filaments. The resulting graph is simpler to describe quantitatively than the original fluorescence pattern.^{1, 2}

A skeleton can support measurements such as:

- filament number, mean length and length distribution;^{3, 4}
- filament density normalized to cell area;
- tortuosity (path length divided by the Euclidean distance between endpoints);¹³
- junction number and branching descriptors;
- global orientation statistics and changes in local direction.

Before skeletonization, CytoOrg applies Fiji's Tubeness command, based on the line-enhancement filter developed by Sato and colleagues.^{5, 6} Tubeness emphasizes curvilinear structures at a selected spatial scale while suppressing much of the non-filamentous background, producing a cleaner binary network for skeletonization.

Sigma is the key Tubeness scale parameter. It determines the filament width preferentially enhanced by the filter. A value that is too small can favor thin noise, whereas a value that is too large can blur or merge nearby structures.⁶ Sigma should therefore match the characteristic filament radius in the image. However, even an appropriately selected sigma cannot compensate for poor image acquisition or severe background noise.⁷

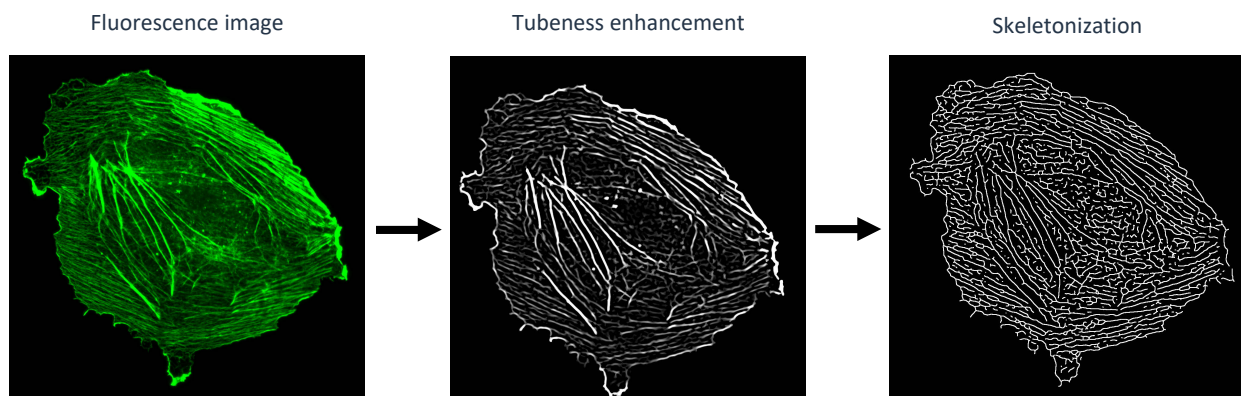


Figure 1. Filament-network extraction by tubeness filtering and skeletonization.

From thickness to network descriptors

A centerline intentionally discards local width. CytoOrg therefore measures thickness before Tubeness filtering and skeletonization when thickness analysis is enabled. After excluding structures below the selected minimum thickness, the script reports the mean, first quartile, median, and third quartile of the local thickness distribution.⁸

Thickness analysis is a result in its own right, but it can also be used to adapt the Tubeness filter dynamically to the characteristic filament thickness of each cell. CytoOrg first measures the mean filament thickness and then sets Tubeness sigma to half of that value (i.e. the mean filament radius):

$$\sigma = \frac{\text{mean filament thickness}}{2}$$

Sigma is therefore recalculated from cell to cell, dynamically adapting the enhancement scale to differences in filament thickness.

Project status and future development

CytoOrg began as a macro of approximately 120 lines. Version 1.0 has grown to 2152 lines and now combines image preparation, thickness analysis, graph-based filament measurements, branching analysis, orientation statistics, visualization, and automated export. At this scale, the project is better suited to a dedicated Fiji plugin architecture than to a progressively larger script.

A plugin version is therefore under development. The planned version will provide its own graphical user interface and native handling of multiple signals or channels, for example a nuclear channel together with microtubules and actin filaments, or additional cytoplasmic and membrane-associated protein signals. Analysis of 3D models may also be considered in future plugin releases. Until then, CytoOrg 1.0 should be used on one exported 2D cytoskeletal channel at a time.

2 Installation

RECOMMENDED ENVIRONMENT Use the current Fiji Stable distribution and install all available updates before running CytoOrg.^{14, 15}

2.1 Requirements

CytoOrg must be run in Fiji, which includes the scripting environment and the AnalyzeSkeleton,⁹ Local Thickness,⁸ and Tubeness commands required by the workflow. OrientationJ must be installed separately.^{10, 16}

The input folder must be writable. CytoOrg creates the results folder next to the input images and, in multi-cell mode, briefly creates a temporary folder in the same location.

2.2 Install OrientationJ

The preferred installation method uses Fiji's built-in updater:^{15, 16}

- **Open the updater.** In Fiji, choose Help > Update...
- **Open the site list.** Click Manage update sites.
- **Enable OrientationJ.** Search and select the BIG-EPFL update site.
- **Apply the change.** Click Close, then Apply changes.
- **Restart Fiji.** The new plugin becomes available after Fiji is restarted.

If the update-site method is unavailable, download OrientationJ_*.jar from the [OrientationJ project page](#), place it in Fiji's plugins folder, and restart Fiji.

2.3 Download and run CytoOrg

- **Download the script.** Get CytoOrg_v1.0.groovy from the [CytoOrg GitHub repository](#).¹⁷
- **Open it in Fiji.** Drag the downloaded file onto the Fiji main window. Fiji opens it in the Script Editor.
- **Select the language.** In the Script Editor, choose Language > Groovy. The *.groovy extension may select it automatically, but it is worth checking.
- **Run the analysis.** Click Run whenever you want to start CytoOrg, then select the folder containing the images to analyze.

SCRIPT, NOT YET A PLUGIN CytoOrg 1.0 does not require permanent installation. Keep the downloaded *.groovy file in a known location and open it in the Script Editor when needed.

3 Accepted inputs

CytoOrg reads all supported image files in the selected input folder, except files whose names begin with `segcell_`. Keep unrelated images out of the analysis folder.

3.1 Core image requirements

Image type. A 2D fluorescence image containing the cytoskeletal signal to be analyzed. The input may be a single optical plane or a 2D projection, such as a maximum-intensity projection (MIP).

Signal channels. One cytoskeletal channel at a time. If the acquisition is multichannel, export the relevant channel as a separate image before analysis.

Supported formats. TIF, TIFF, PNG, JPG, and JPEG; extension matching is case-insensitive.

Field of view. The entire cell should be visible. Avoid cropped cells unless the experimental design explicitly requires them.

Spatial calibration. Know the pixel size in micrometers per pixel. CytoOrg applies the value entered under General settings.

3.2 Single-cell images

The simplest input contains one isolated cell per image, with every signal outside that cell removed. In this mode, CytoOrg derives the cell outline from the non-zero pixels, fills holes, and uses the resulting region to calculate cell area, roundness, and solidity¹¹ before analyzing the cytoskeleton.

3.3 Preparing a single-cell image in Fiji

The following workflow creates a cell ROI from a temporary duplicate and applies it to the original image. Menu names are written exactly as they appear in Fiji.

Duplicate the image. Choose Image > Duplicate... and click OK. Keep the original image open.

Convert the duplicate to 8-bit. Choose Image > Type > 8-bit.

Threshold the cell. Choose Image > Adjust > Threshold.... Li and Otsu are useful starting methods. Click Auto for an initial threshold, then inspect the full cell boundary. If weak peripheral regions are excluded, lower the threshold slightly until the outline is complete, accepting only limited background noise. To compare thresholding methods, choose Image > Auto Threshold and select Try all.

Correct the outline if needed (optional). Choose Edit > Options > Colors... and set Foreground to white. With the line tool, draw a short selection across each remaining gap and choose Edit > Draw. Repeat only where needed to obtain a closed cell boundary before filling the interior.

Fill the interior. Choose Process > Binary > Fill Holes.

Create a cell ROI. Choose Analyze > Analyze Particles.... Enable Add to Manager and set a minimum size that removes debris, such as 500-Infinity with Pixel units enabled.

Inspect the detected ROI. In the ROI Manager, select the ROI corresponding to the cell and confirm that it follows the intended cell boundary.

Return to the original image. Activate the original image and select the same cell ROI in the ROI Manager.

Set a black background. Choose Edit > Options > Colors... and set Background to black so that Clear Outside assigns intensity 0 to removed pixels.

Remove everything outside the cell. Choose Edit > Clear Outside

Save the prepared image. Choose File > Save As > Tiff....

Reset before the next cell. Clear the ROI Manager and close the working images before repeating the procedure with the next sample.

THRESHOLDING IS DATA-DEPENDENT No thresholding method is optimal for every dataset. Compare methods on representative images, select the rule that consistently captures the complete cell without joining neighboring cells or retaining artifacts.

3.4 Multi-cell images with segmentation masks

CytoOrg can analyze several cells in one image when Use segcell masks for multiple cells is enabled. Each original image must have a matching binary cell mask in the same input folder. The mask filename must begin with segcell_ followed by the complete original filename, including its extension.

For example:

Original image: field_01.tif

Cell mask: segcell_field_01.tif

This multi-cell workflow was inspired by the segmentation-mask approach used in CytoSpectre.^{12, 18}

A valid segmentation mask must meet all of the following conditions:

Exact size. The mask and original image must have identical width and height. CytoOrg checks these dimensions before analysis.

One component per cell. Each cell must be a separate connected white object. Cells must not touch or be linked, including by narrow bridges or corner contacts.

No debris. Remove all unwanted white objects. Every detected white component is treated as a cell because the script uses the range 0-Infinity.

One mask per image. When multi-cell mode is enabled, every original image in the folder must have a matching segcell_mask. The analysis stops if any mask is missing.

What CytoOrg does with the mask

The script validates that each original image has a mask with matching width and height.

Fiji Analyze Particles identifies every white connected component and adds the corresponding cell ROI to the ROI Manager.

CytoOrg records cell area, roundness, and solidity from the mask.

A temporary folder named `.cytoorg_temp_<timestamp>` is created inside the input folder.

For each ROI, CytoOrg saves a temporary TIFF copy of the original image with all pixels outside that cell set to black, then runs the same analysis used for a single-cell image.

A control image named `ROI_overlay_<original name>.tif` is saved with yellow ROI outlines and the cell numbers used in output names and result rows.

The temporary single-cell images and the temporary folder are deleted when processing finishes normally.

CELL NUMBERING Fiji Analyze Particles numbers disconnected objects from top to bottom and, within the same row, from left to right.

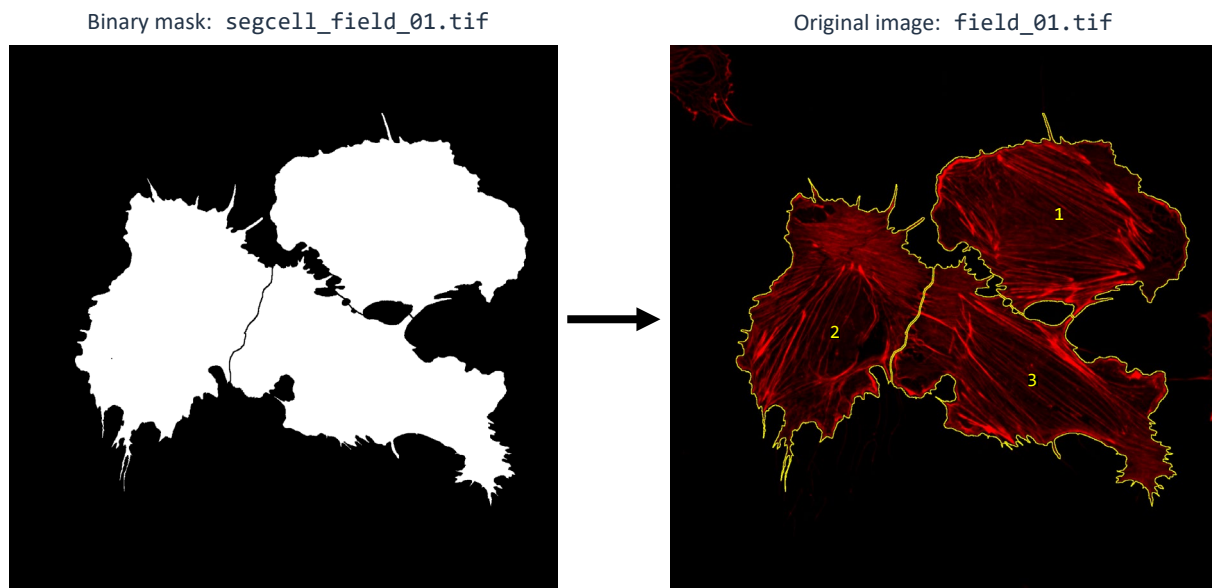


Figure 2. `segcell_masks` define the cells included in multi-cell analysis (yellow outlines).

IF A RUN IS INTERRUPTED Images and plots already written remain available, and the open results table can be copied manually or saved as a `.csv` file. The `.cytoorg_temp_<timestamp>` folder may also remain and can be safely deleted. After checking the saved outputs, rerun only the remaining images or remove previously processed cells from the masks.

For more information, see Section 7, Troubleshooting and feedback.

4 Settings

After the input folder is selected, CytoOrg opens the main settings dialog. This dialog contains the controls required for a standard analysis. The three *Advanced ... settings* checkboxes open further dialog windows after the main dialog is confirmed, allowing the corresponding default values and optional outputs to be changed.

CHOOSE SETTINGS BEFORE ANALYZING THE FULL DATASET Optimize the settings on a small, representative group of cells that includes weak, strong, thin, thick, sparse, and dense examples.² Once the processed skeletons are biologically plausible, keep the same settings for all images that will be compared. A parameter should not be adjusted cell by cell unless the workflow explicitly defines an adaptive rule, such as automatic Tubeness sigma.

ENHANCED MAPS ARE FOR VISUALIZATION ONLY Enhanced skeleton, branching, and orientation maps are thickened copies created only to make one-pixel-wide structures easier to see. They are never used for measurements.

4.1 General settings

Use segcell masks for multiple cells. Default: off. Enable this option when an original field contains more than one cell and a matching binary mask named `segcell_<complete image filename>` is provided.

Pixel size. Default: 0.055 $\mu\text{m}/\text{pixel}$. Enter the physical size represented by one pixel. CytoOrg uses this calibration to report lengths in micrometers and areas in square micrometers, and to convert physical thresholds into pixel units.

Filament definition. This setting determines what CytoOrg treats as one filament when calculating filament number, length, tortuosity, density, and global orientation statistics. It does not change the one-pixel skeleton itself.

- **(A) Segments between junctions/endpoints.** Default. Every graph edge is one filament: a path from one endpoint or junction to the next endpoint or junction. This definition is appropriate when the biological question concerns the local pieces that form a branched network. Highly branched networks will contain many short segments because each junction divides a path into separate filaments.
- **(B) Main filament (longest shortest path).** For each filament network, CytoOrg uses an all-pairs shortest-path (APSP) calculation: it finds the shortest route between every pair of endpoints and selects the longest of these routes as the main filament.¹³ The remaining segments are side branches. This definition is useful when the continuous backbone is the structure of interest and generally produces fewer, longer main filaments than the segment definition.

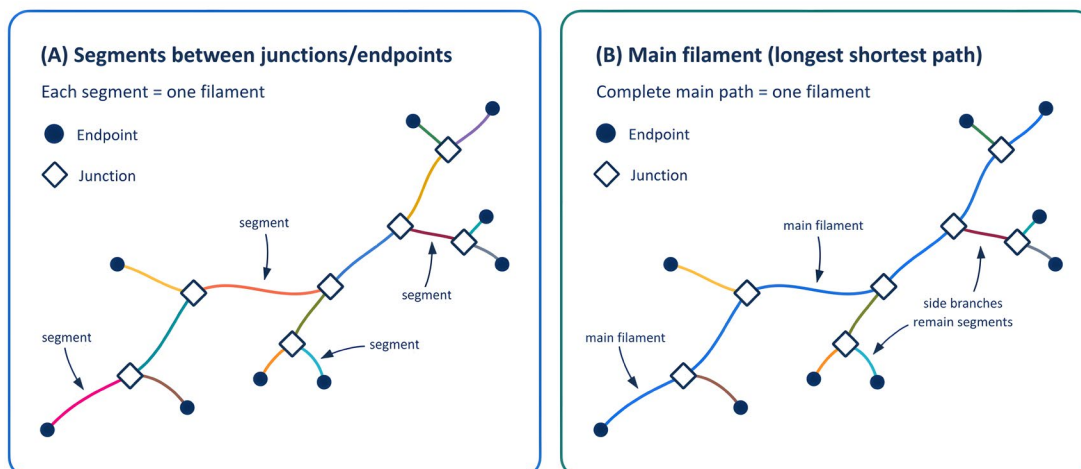


Figure 3. Comparison of the segment and main-path filament definitions in CytoOrg.

Output folder name. The folder is created inside the selected input folder. Changing its name is useful for organizing separate analyses or repeating an analysis without overwriting earlier results.

4.2 Thickness settings

CytoOrg uses Fiji Local Thickness to assign a local diameter to foreground pixels, summarizes the retained values, and saves a color-coded thickness map.

Thickness settings

Measure filament thickness. Default: on. When enabled, CytoOrg measures local thickness, reports its mean and quartiles, and saves a thickness map for every cell. Disable it when only network geometry is required or when the signal cannot support a meaningful width measurement. If it is disabled, automatic Tubeness sigma cannot be used.

Minimum thickness. Default: 0.20 μm . Local-thickness values at or below this cutoff are excluded from the thickness distribution and set to black in the saved thickness map.

Advanced thickness settings

Intensity threshold (0 = automatic). Default: 0. This threshold removes low-intensity intracellular background before Local Thickness is calculated, so the measurement focuses on the structures of interest. With 0, CytoOrg applies the Otsu method automatically. A fixed value from 1 to 255 can instead be used when images have consistent intensity scaling; increasing it retains only brighter structures, whereas decreasing it includes weaker signal and more background.

Thickness-map color scale. Default: minimum 0 μm ; maximum seven times the minimum thickness. These limits change only the LUT color mapping, not the measured values. Out-of-range values are visually saturated at the corresponding scale limit. Use the same limits when maps must be visually comparable. The color bar for the applied LUT is available in the CytoOrg GitHub repository.¹⁷

4.3 Skeleton settings

Skeleton analysis is always performed. CytoOrg first applies Tubeness, converts the enhanced signal to a binary mask and skeletonizes the mask. AnalyzeSkeleton then converts the skeleton into connected graphs¹³ and provides the data used to calculate filament and branching descriptors.

Skeleton settings

Tubeness sigma. Default: 0.30 μm . Sigma sets the physical scale preferentially enhanced by the Tubeness filter. A larger value favors wider, smoother structures and suppresses smaller detail, but it can broaden or merge nearby filaments and create false connections.⁶ A smaller value favors thin structures, but it can fragment filaments or enhance fine noise. Select a value close to the characteristic filament radius visible in representative images.

Estimate sigma from filament thickness. Default: off. When enabled, CytoOrg calculates a separate Tubeness sigma for every cell as one half of that cell's retained mean thickness. This is useful when filament width varies substantially among cells and a single scale would over-enhance some cells while under-enhancing others. Leave it off when a common filtering scale is required for strict comparability, or when thickness measurement is unreliable. This option overrides the manual Tubeness sigma and requires thickness measurement.

Minimum length. Default: 0.20 μm . This is the main lower size cutoff for skeleton analysis. CytoOrg uses it during cleanup of very small skeleton components and later includes a filament in the main summary only when its path length is greater than the cutoff and its endpoint distance is greater than zero. The raw grouped and individual-filament data described below are deliberately saved before this filtering.

Advanced skeleton settings

CytoOrg provides the following AnalyzeSkeleton pruning options, which selectively remove parts of the skeleton before measurement.¹³

- **Loop pruning method.** Default: None. With None, closed loops are preserved; use this when loops may be real network structures or when there is no validated reason to remove them. With Shortest branch, AnalyzeSkeleton breaks each detected loop by cutting its shortest participating branch. This can remove false closures caused by segmentation or projection, but it also breaks genuine loops. If enabled, all downstream measurements use the pruned skeleton.
- **Prune branches ending at end-points.** Default: off. When enabled, AnalyzeSkeleton removes terminal branches that end at an endpoint. This option is generally unsuitable for cytoskeletal analysis, as it removes genuine filament ends along with any artifacts. Leave it disabled unless the aim is specifically to isolate the network backbone rather than characterize the full filament network. If enabled, all downstream measurements use the pruned skeleton.

Group filaments by length. Default: off. When enabled, CytoOrg adds the number and density of short, intermediate, and long filaments to the main Results.csv table. This classification does not alter which filaments contribute to the overall measurements; it adds parallel summaries for the three classes.

Short/intermediate threshold (T1). Default: nine times the minimum length. Retained filaments with $\text{length} < T1$ are classified as short. Increasing T1 moves more filaments into the short class; decreasing it moves more into the intermediate class. T1 must be at least the minimum length and lower than T2.

Intermediate/long threshold (T2). Default: seventeen times the minimum length. Retained filaments with $T1 \leq \text{length} < T2$ are intermediate, whereas filaments with $\text{length} \geq T2$ are long. Increasing T2 expands the intermediate class and reduces the long class; decreasing it has the opposite effect.

Export individual filament data. Default: off. Saves one `Single_filament_<image>.csv` file for each analyzed cell or image, with one row per filament and its skeleton ID, path length, Euclidean endpoint distance, and endpoint coordinates. Unlike `Results.csv`, which contains one summary row per cell, these files preserve every filament measurement. They can therefore be combined across cells or experimental groups to calculate population-level statistics, such as the standard deviation of all filament lengths. The table is unfiltered and is intended for custom downstream analysis.

Export grouped filament data. Default: off. Saves one `Grouped_filament_<image>.csv` file for each analyzed cell or image. Unlike the individual-filament table, each row represents one connected filament network and summarizes its graph-level structure.¹³ The table is unfiltered and is intended for custom downstream analysis.

Save enhanced skeleton. Default: on. The standard one-pixel skeleton is always saved as `Skeleton_<image>`. This option additionally saves `Improved_Skeleton_<image>`, in which the lines are thickened for easier visual inspection.

Branching settings are also grouped under skeleton analysis because the method builds on the longest shortest path map provided by `AnalyzeSkeleton`.¹³

Analyze filament branching. Default: on. When enabled, CytoOrg identifies main-path and side-branch structures and reports the fraction of branched filaments, the side-branch/all filament length ratio, and the rebranching rate. Disable it when these descriptors are not relevant. When it is disabled, both branching-map options are automatically disabled.

Save branching map. Default: off. Saves the one-pixel-wide branching map, with main filaments in cyan and side branches in yellow. The color key is available in the CytoOrg GitHub repository.¹⁷ Saving the map does not affect the calculations.

Save enhanced branching map. Default: on. Saves `Improved_Branching_<image>`, in which the colored lines are thickened for easier visual inspection. The enhanced map can be saved without saving the standard map, and this is the default behavior meant to provide more readable images for presentations, posters, and publication figures.

4.4 Orientation settings

Analyze filament orientation. Default: on. Calculates global filament-orientation statistics and uses OrientationJ to measure local orientation changes along the skeleton. Disable it when orientation is not relevant; this also shortens the analysis time.

AXIAL ORIENTATION A filament is undirected: reversing its endpoints does not change its orientation. Angles therefore range from 0° to 180° excluded, and 0° is equivalent to 180°. ¹⁸

Advanced orientation settings

Exclude side branches from orientation statistics. Default: off. When off, main paths and side branches both contribute. When on with the main-filament definition, only complete longest paths are retained for global orientation statistics. When on with the segment definition, only the segments that belong to those longest paths are retained. This setting does not remove side branches from other analysis, including OrientationJ local direction-change analysis.

Maximum tortuosity included. Default: 1.5; the minimum valid value is 1. A value close to 1 accepts only nearly straight filaments, for which a single endpoint-to-endpoint angle is a good description. Increasing the limit includes more curved filaments, but their endpoint angle may hide substantial local bending. Decreasing it makes the global orientation estimate more conservative and often reduces the number of accepted filaments. This filter affects global orientation statistics and the angular distribution plot only; it does not change the mean tortuosity or the OrientationJ direction-change measurement.

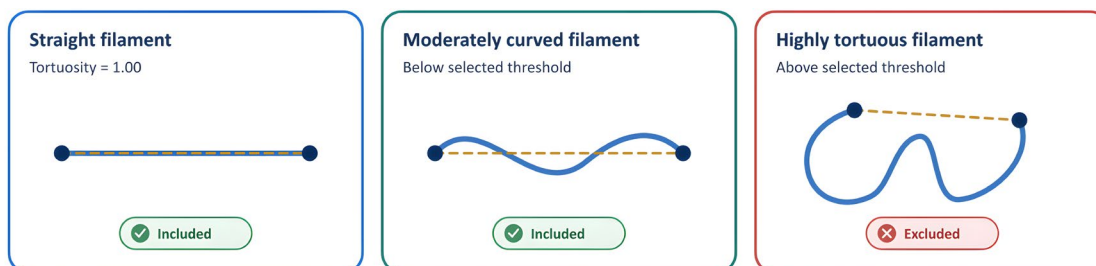


Figure 4. Effect of the maximum-tortuosity filter on global orientation analysis.

Weight orientation statistics by filament length. Default: on. When enabled, a filament contributes in proportion to its length, so a small number of long filaments can dominate the mean orientation and angular distribution. When disabled, every accepted filament contributes equally, so numerous short filaments can dominate. Length weighting is often appropriate when the aim is to describe how much of the network is aligned in each direction; equal weighting is appropriate when every filament is considered one biological observation. Keep the choice fixed across comparisons.

Save angular distribution plot. Default: on. Saves Angular_Distribution_<image>.tif, a polar representation of the same accepted filaments and weights used for global orientation statistics. Because filament orientation is axial, the 0-180° distribution is repeated over 180-360° for a complete circular display. Turning off the plot does not disable the numerical orientation results.

Angular bin width. Default: 5°. It must be an integer divisor of 180 between 1° and 90°, for example 1°, 2°, 3°, 5°, or 10°. A smaller bin width provides finer angular resolution but produces a more variable, spiky distribution when few filaments are available. A larger bin width gives a smoother and more robust overview but can merge nearby orientation peaks.

Radial-axis maximum (0 = automatic). Default: 0. With 0, each plot is scaled to its own highest frequency, making its shape easy to inspect. A value from 1% to 100% fixes the outer circle at that frequency for every plot; frequencies above it are clipped. This setting changes only the plot scale, not the numerical orientation results.

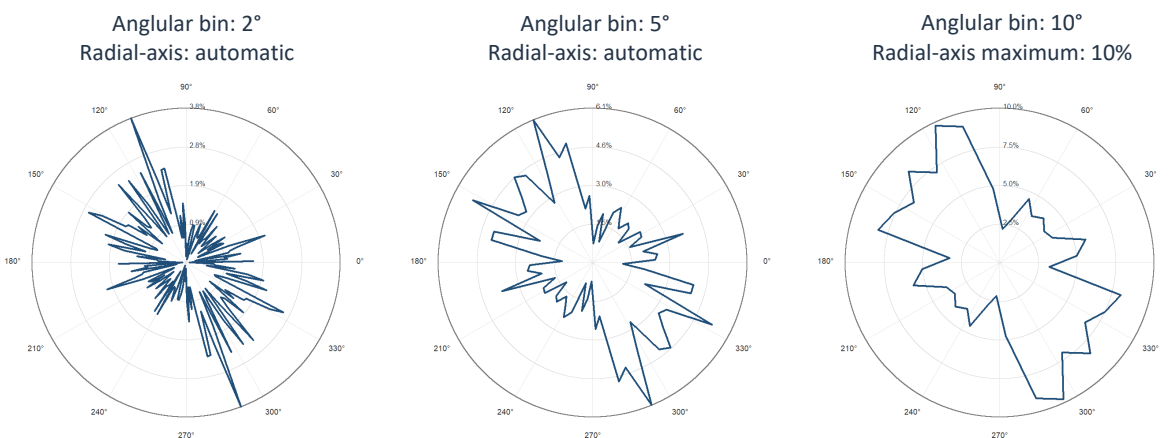


Figure 5. Effect of angular bin width and radial-axis maximum on plots generated from the same data.

OrientationJ estimates the local direction of the skeleton.¹⁰ Its settings affect the orientation map and the Direction Changes per μm result and should be kept constant within a dataset.

Orientation window sigma. Default: 2 pixels. Sigma defines the size of the local neighborhood used to estimate direction. A smaller value follows fine bends but is more sensitive to pixelation and noise; a larger value produces a smoother orientation map but can average sharp bends or nearby filaments.

Gradient method. Default: Cubic Spline. This setting determines how OrientationJ calculates local intensity changes. Cubic Spline is the general-purpose default. For details on the other methods, see the OrientationJ documentation.¹⁶

Minimum direction change (degrees). Default: 45°; allowed range: 1-90°. Only local orientation differences at least this large are counted. A higher value counts only sharper bends; a lower value detects subtler changes but is more sensitive to pixelation and local fluctuations.

Save orientation map. Default: off. Saves Orientation_<image>, the one-pixel-wide OrientationJ map in which hue represents local axial orientation. The corresponding color bar is available in the CytoOrg GitHub repository.¹⁷ Saving the map does not affect the analysis.

Save enhanced orientation map. Default: on. Saves Improved_Orientation_<image>, in which the colored lines are thickened for easier visual inspection.

5 Results

CytoOrg reports one row per analyzed cell in the results table, which remains open in Fiji and can be copied and pasted directly into a spreadsheet. The same table is saved as `Results.csv`. If Excel places the CSV content in a single column, select that column and choose Data > Text to Columns > Delimited > Comma. Thickness, branching, and orientation columns are included only when the corresponding analyses are enabled.

ZERO AND NAN ARE NOT INTERCHANGEABLE A value of 0 is a defined result and normally means that the measured feature was absent. NaN means that the result was mathematically undefined, most often because no filament passed the relevant filters or because a denominator was zero. A non-finite value should be treated as missing data, not converted automatically to zero.

5.1 Cell morphology results

These values describe the two-dimensional projected cell outline. Cell area is also used to normalize cytoskeletal measurements reported per unit area.

Cell Area (μm^2). The area enclosed by the cell mask, calculated as the number of foreground pixels multiplied by the calibrated pixel area.

Cell Roundness. A measure of cell elongation calculated by ImageJ as

$$\text{Roundness} = \frac{4 \cdot A_{\text{cell}}}{\pi \cdot M^2}$$

where A_{cell} is cell area and M is the major axis of the best-fitting ellipse. Range: 0 – 1. Values close to 1 indicate little or no elongation; values approaching 0 represent increasingly elongated cells.

Cell Solidity. The ratio

$$\text{Solidity} = \frac{A_{\text{cell}}}{A_{\text{convex}}}$$

where A_{cell} is cell area and A_{convex} is the area of its convex hull, the smallest convex outline containing the cell.¹⁸ Range: 0 – 1. A value of 1 represents a completely convex outline. Lower values indicate that less of the convex hull is covered by the cell because its boundary contains concavities, indentations, or separated protrusions.

Multi-cell ROI control output

ROI_overlay_<original image>.tif. A copy of the original multi-cell image containing yellow, 2-pixel-wide ROI outlines and bold yellow cell numbers. The numbering corresponds exactly to the suffixes `_cell1`, `_cell2`, and so forth and therefore links each object in the field to its row in `Results.csv`. This file is a quality-control record only: it is created after the mask objects are identified and is not used for any measurement. It is produced only in multi-cell analysis.

5.2 Thickness results

Thickness results are present only when *Measure filament thickness* is enabled. Local Thickness assigns a diameter to each foreground pixel from the largest circle that fits entirely inside the thresholded structure and contains that pixel.⁸ CytoOrg retains only values strictly greater than the selected minimum thickness. The statistics below summarize all retained pixels, so a broad structure contributes more observations than a narrow structure of the same length. They are pixel-weighted local-diameter statistics, not an average of one value per filament.

Mean Thickness (μm). The arithmetic mean of all retained local-diameter values: the sum of the retained pixel values divided by their number, followed by conversion from pixels to micrometers.

Q1 Thickness (μm). The first quartile of the retained local-diameter distribution. Approximately one quarter of retained pixels have a thickness at or below this value and three quarters at or above it.

Median Thickness (μm). The central retained local-diameter value after sorting. For an odd number of pixels, it is the middle value; for an even number, it is the mean of the two middle values. Half of the retained pixels have thicknesses at or below the median and half at or above it. It is less sensitive than the mean to a large number of very thin or very thick regions.

Q3 Thickness (μm). The third quartile of the retained local-diameter distribution. Approximately three quarters of retained pixels have a thickness at or below this value and one quarter at or above it.

WHY REPORT QUANTILES Q1, the median, and Q3 describe the distribution of local thickness values more completely than the mean alone, showing its lower, central, and upper portions.

Saved thickness map

Thickness_<image>.tif. A color-coded map of the local diameters measured before Tubeness filtering and skeletonization. Each foreground pixel retained after intensity thresholding is colored according to its local diameter; values at or below the minimum-thickness cutoff are black. The selected color-scale limits affect only the appearance of the saved map.

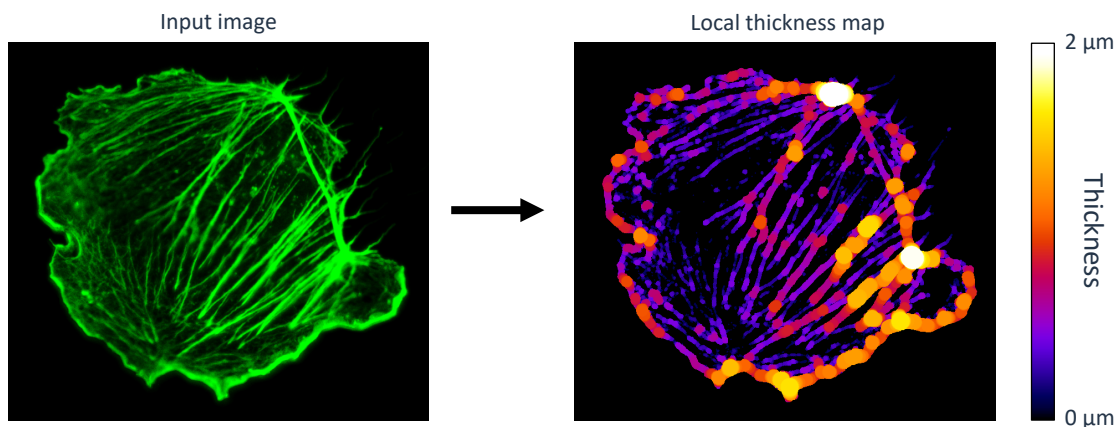


Figure 6. Local thickness mapping of an actin immunofluorescence image.

5.3 Skeleton results

Skeleton results are calculated from the processed, optionally pruned skeleton using the selected filament definition. The exception is Cytoskeleton Area, which is measured before skeletonization from the filamentous structures recognized by the Tubeness filter. Unless stated otherwise, a filament is included only when its path length is greater than the minimum length and its Euclidean endpoint distance is greater than 0 (and its intensity is not zero, to exclude background artifacts that AnalyzeSkeleton incorrectly identifies as filaments).

Cytoskeleton Area (μm^2). The calibrated area of the binary Tubeness-positive foreground immediately before skeletonization. A larger value represents a greater two-dimensional area classified as cytoskeleton, although it is also affected by Tubeness sigma, signal intensity, and thresholding.

A_{csk}/A_{cell} (%). The cytoskeleton-to-cell area percentage, calculated as

$$A_{\text{csk}}/A_{\text{cell}} (\%) = 100 \cdot \frac{\text{Cytoskeleton Area}}{\text{Cell Area}}$$

Higher values indicate that Tubeness-positive cytoskeletal structures occupy a larger proportion of the cell area, reflecting greater cytoskeletal coverage and apparent density.

N. of Filaments. The number of valid filament records produced by the selected definition after the minimum-length and non-zero-endpoint-distance filters. With the segment definition, each path between two graph vertices is counted separately. With the main-filament definition, each longest shortest path is one main filament, while side branches are counted separately as paths between consecutive junctions and/or endpoints.

N. of Junctions. The sum of AnalyzeSkeleton junctions across connected skeleton graphs whose average branch length is strictly greater than the minimum filament length. A junction is a graph location where paths meet; adjacent junction pixels are treated as one junction cluster by AnalyzeSkeleton.¹³

Junction/Filaments. The average number of junctions per retained filament and therefore a direct measure of branching. A value of 0 means that no junction is present; higher values indicate more branches per filament and therefore greater branching.

Mean filament length (μm). The sum of all valid filament lengths divided by N. of Filaments. Higher values indicate longer filaments under the chosen definition.

Std Dev filament length (μm). The sample standard deviation of valid filament lengths, calculated with denominator N - 1. It is 0 when there is one retained filament or all retained lengths are identical, and it increases as filament lengths become more heterogeneous.

Total filament length (μm). The sum of the lengths of all valid filament records. This measure generally increases with cell size.

Maximum filament length (μm). The greatest length among the valid filament records.

Mean Tortuosity. Compares the total filament path length with the total straight-line distance between filament endpoints:

$$T_{\text{mean}} = \frac{\sum L_i}{\sum D_i}$$

L is the filament path length and D is its Euclidean endpoint distance. Range: 1 or greater. A value of 1 represents straight filaments; higher values represent increasingly curved or winding filaments. Because the totals are used, longer filaments contribute more strongly.

Filament Density per 100 μm^2 . A count density calculated as

$$\text{Density}_{\text{filaments}} = 100 \cdot \frac{N_{\text{filaments}}}{A_{\text{cell}}}$$

Larger values mean that more filaments are present per 100 μm^2 of cell area, according to the selected filament definition. This is not a length density: many short filaments can produce a high value even when their total length is modest.

Optional length-class results. These six columns are present only when *Group filaments by length* is enabled. Every valid filament enters one class. With thresholds $T1$ and $T2$, short filaments satisfy $L < T1$, intermediate filaments satisfy $T1 \leq L < T2$, and long filaments satisfy $L \geq T2$. Thus, despite the greater-than sign used in the long-column header, a filament exactly equal to $T2$ is included in the long class.

Number of filaments in each length class. Counts of short ($L < T1$), intermediate ($T1 \leq L < T2$), and long ($L \geq T2$) filaments.

Length-class filament density. The short, intermediate, and long filament counts are each normalized to cell area with the same equation:

$$\text{Density}_{\text{class}} = 100 \cdot \frac{N_{\text{class}}}{A_{\text{cell}}}$$

where class is short, intermediate, or long. The result is the number of filaments in that class per 100 μm^2 of cell area.

DENSITY SCALE Densities can be rescaled to any reference area, such as 1 or 1,000 μm^2 . CytoOrg reports values per 100 μm^2 to provide an intuitive scale for typical cell sizes.

Raw filament tables (see also section 4.3)

Single_filament_<image>.csv. The unfiltered filament-level table generated from the processed and optionally pruned skeleton using the selected filament definition. Each row contains the skeleton identifier, filament length, Euclidean endpoint distance, and the x, y, and z coordinates of both endpoints; z is 0 for a two-dimensional image. Average filament intensity is also reported, but a known AnalyzeSkeleton bug may return 0 instead of the expected 255 for the binary skeleton.¹³ These rows remain in the raw CSV but are excluded from filtered statistics.

Grouped_filament_<image>.csv. The unfiltered AnalyzeSkeleton table, with one row per connected skeleton. Skeleton identifies the object; # Branches counts its graph edges; # Junctions counts locations where branches meet; # End-point voxels, # Junction voxels, and # Slab voxels count the three skeleton-pixel classes; Average Branch Length and Maximum Branch Length summarize edge lengths; # Triple points and # Quadruple points count junctions with three or four branches; Longest Shortest Path gives the main-path length; and spx, spy, and spz give the coordinates of its starting point.¹³ These results always follow the segment definition (may be corrected in a future release).

Saved skeleton maps

Skeleton_<image>.tif. The binary, one-pixel-wide centerline used for all graph, branching, and local-orientation analyses. The background is black and skeleton pixels are white. It is created after small skeleton components are removed and, when loop or end-branch pruning is enabled, it is overwritten with the pruned version returned by AnalyzeSkeleton.

Improved_Skeleton_<image>.tif. A display copy of the processed skeleton in which a 3-pixel-radius maximum filter expands the white centerlines, making them easier to inspect at normal document scale.

5.4 Branching results

Branching results are present only when *Analyze branching* is enabled. AnalyzeSkeleton identifies the longest shortest path through each connected skeleton graph.¹³ CytoOrg treats these paths as the main network and classifies the remaining pixels as side branches. The following metrics describe how much structure lies outside the main paths and how often side-branch structures subdivide.

Fraction of Branched Filaments (%). A component-count ratio calculated as

$$\text{Fraction}_{\text{side}}(\%) = 100 \cdot \frac{N_{\text{side}}}{N_{\text{side}} + N_{\text{main}}}$$

N_{side} is the number of side-branch filaments and N_{main} is the number of main filaments identified as longest shortest paths. A higher percentage means that more filaments are classified as side branches rather than main filaments, reflecting a more extensively branched cytoskeletal network.

Branches/Filaments length ratio. The number of side-branch skeleton pixels divided by the total number of classified main-path and side-branch skeleton pixels

$$\text{Ratio}_{\text{length}} = \frac{L_{\text{side}}}{L_{\text{side}} + L_{\text{main}}}$$

Range: 0 – 1. A value of 0 means that none of the total filament length is assigned to side branches; a value approaching 1 means that an increasingly large fraction of the total length belongs to side branches rather than main filaments. Unlike the fraction of branched filaments, this ratio reflects branch length rather than branch number. A network with many short side branches can therefore have a similar ratio to one with fewer, longer side branches.

Rebranching rate. The estimated number of side-branch segments ($N_{\text{segments}} - N_{\text{main paths}}$) divided by the number of connected side-branch objects ($N_{\text{side filaments}}$). Each object is a separate group of one or more connected side branches remaining after the longest shortest paths are removed. CytoOrg estimates the number of side-branch segments as the total number of valid AnalyzeSkeleton segments (i.e. segment definition) minus the number of longest shortest paths (i.e. main filament definition).

$$\text{Rate}_{\text{rebranching}} = \frac{N_{\text{segments}} - N_{\text{main paths}}}{N_{\text{side filaments}}}$$

A value near 1 indicates mostly simple, unbranched side filaments; higher values indicate that side branches themselves divide into further branches, reflecting greater network complexity.

Saved branching maps

Branching_<image>.tif. A color classification of the processed skeleton. Cyan marks the longest shortest path assigned to the main network of each connected skeleton, yellow marks pixels classified as side branches, and black marks the background.

Improved_Branching_<image>.tif. A copy of the branching map in which the colored lines are thickened for easier visual inspection. It is not used for measurements.

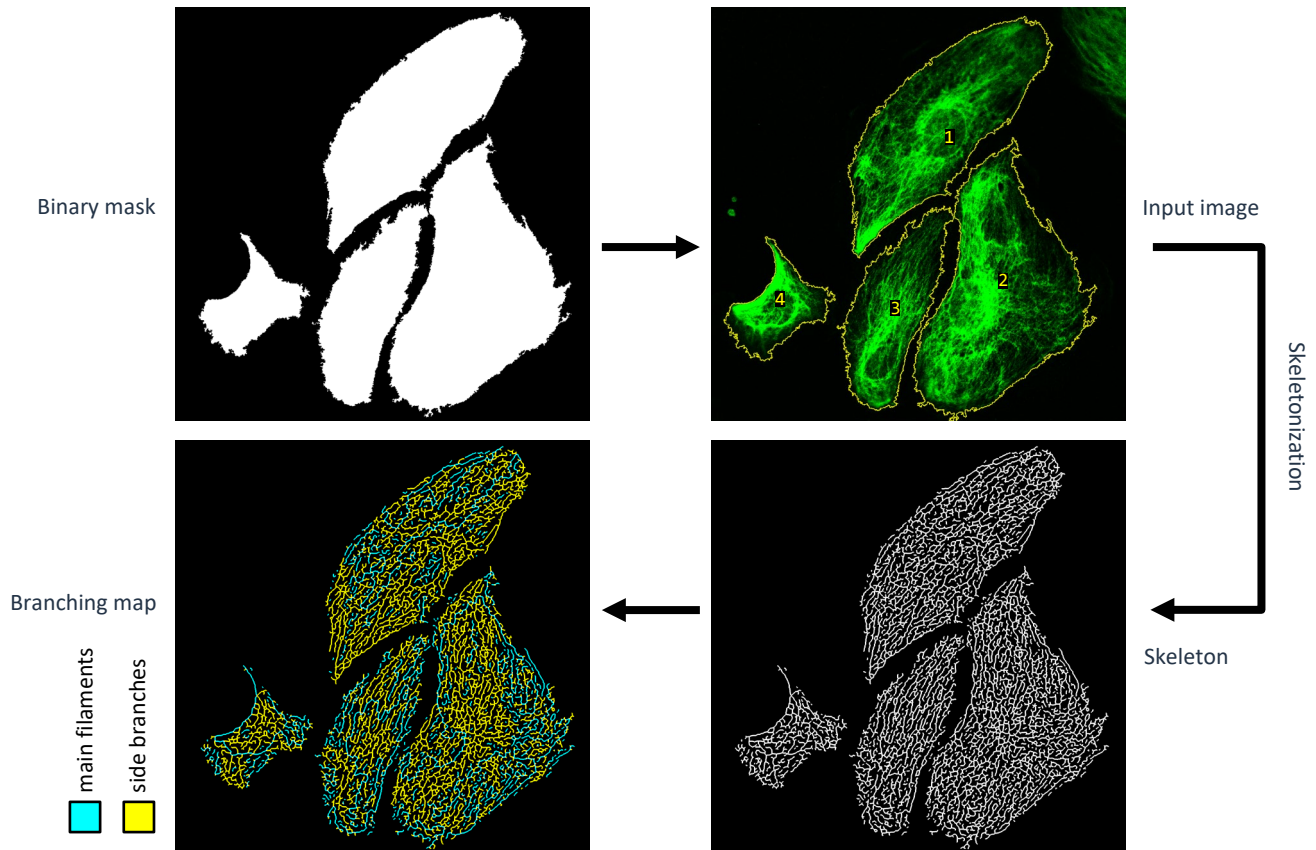


Figure 7. Network branching assessment in a multi-cell analysis of vimentin immunofluorescence.

5.5 Orientation results

CytoOrg evaluates orientation at two scales. Global orientation statistics and the angular distribution describe the endpoint-to-endpoint axis of each accepted filament. Local orientation analysis uses OrientationJ to follow direction changes along the processed skeleton.

For an accepted filament, the endpoint direction is

$$\theta = \text{atan2}(y_1 - y_2, x_2 - x_1)$$

normalized to $0^\circ \leq \theta < 180^\circ$. This is axial orientation: a filament pointing in opposite directions has the same orientation, so 0° and 180° represent the same horizontal axis, while 90° represents a vertical axis.

FILAMENT WEIGHTING In the following equations, θ_i is the orientation of filament i and w_i is its weight. When length weighting is enabled, $w_i = \text{Length}_i$, so longer filaments contribute more strongly. When disabled, $w_i = 1$, so every filament contributes equally. Accordingly, $\sum w_i$ is either the total filament length or the number of filaments included in the orientation analysis.

Mean Orientation (Degrees). The weighted axial circular mean.¹⁸ CytoOrg averages $\cos(2\theta)$ and $\sin(2\theta)$, calculates half of atan2 of those averages, and normalizes the result to $[0^\circ, 180^\circ)$

$$\mu = \frac{1}{2} \cdot \text{atan2}(\sum w_i \cdot \sin(2\theta_i), \sum w_i \cdot \cos(2\theta_i))$$

Range: 0° inclusive to 180° exclusive, with 0° and 180° being equivalent. When the distribution is isotropic, contributions from different directions cancel and the orientation order parameter approaches 0; in this case, the mean direction is unstable and should not be interpreted alone.¹²

Orientation Order Parameter. A measure of directional variability and alignment.¹ It combines the weighted averages of $\cos(2\theta)$ and $\sin(2\theta)$:

$$\text{OOP} = \frac{\sqrt{(\sum w_i \cdot \cos(2\theta_i))^2 + (\sum w_i \cdot \sin(2\theta_i))^2}}{\sum w_i}$$

Range: 0 – 1. When filaments share the same orientation, the trigonometric terms reinforce one another and OOP approaches 1, indicating an anisotropic, strongly aligned cytoskeleton. When filament orientations are broadly distributed, the terms cancel and OOP approaches 0, indicating an isotropic cytoskeleton without a dominant axis.

Circular Skewness. A measure of the symmetry of the angular distribution around the mean orientation:¹⁸

$$\text{Skewness} = \frac{\sum w_i \cdot \sin[4 \cdot (\theta_i - \mu)]}{\sum w_i}$$

Range: -1 to 1. Values near 0 indicate that filament orientations are distributed symmetrically around the mean. A larger absolute value indicates that more filament weight lies on one angular side of the mean; the sign indicates the direction of this asymmetry.

Circular Kurtosis. A measure of the peakedness of the angular distribution:¹⁸

$$\text{Kurtosis} = \frac{\sum w_i \cdot \cos[4 \cdot (\theta_i - \mu)]}{\sum w_i}$$

Range: -1 to 1. Absolute values close to 1 indicate a strongly peaked distribution, with filaments concentrated around a limited number of orientation angles. Values closer to 0 indicate that filament orientations are spread more smoothly across many angles. Kurtosis describes peakedness, not alignment by itself, and should therefore be interpreted together with the Orientation Order Parameter.

Angular distribution plot

Angular_Distribution_<image>.tif. A 900 × 900 pixel polar plot built from the same accepted filaments and weights used for the global orientation statistics. CytoOrg assigns each axial angle in [0°, 180°) to a bin of the selected width and expresses the bin weight as

$$\text{Frequency}_{\text{bin}}(\%) = 100 \cdot \frac{\sum_{i:\theta_i \in \text{bin}} w_i}{\sum w_i}$$

The 0-180° distribution is mirrored over 180-360° because filament orientation is axial. The two semicircles therefore contain the same information, and only one semicircle needs to be interpreted. Points are placed at bin centers and connected by a dark-blue line. The four rings represent 25%, 50%, 75%, and 100% of the maximum frequency, while spokes and angle labels are placed every 30°.

Saved orientation maps

Orientation_<image>.tif. The one-pixel-wide OrientationJ map calculated from the processed skeleton. Hue represents local axial orientation,¹⁶ saturation and brightness are constant, and the background is dark. The map is used to calculate Direction Changes per μm.

Improved_Orientation_<image>.tif. A copy of the orientation map in which the colored lines are thickened for easier visual inspection. It is not used for measurements.

Direction Changes per μm. OrientationJ divides the skeleton into connected components whose pixels share a similar local direction. A component can represent an entire straight filament or a portion of a filament whose direction differs from its local surroundings. CytoOrg subtracts the number of components already present in the original skeleton (i.e. the number of filament networks), so no direction changes are counted for a filament that remains straight. The remaining changes are normalized by total filament length:

$$\text{Direction changes}/\mu\text{m} = \frac{N_{\text{orientation components}} - N_{\text{skeleton components}}}{\text{Length}_{\text{total}}}$$

Higher values indicate more frequent changes in local filament direction per micrometer. Biologically, this reflects a cytoskeletal network with more frequent bends and turns, whereas lower values indicate straighter filament trajectories.

INTERPRET GLOBAL AND LOCAL ORIENTATION SEPARATELY Mean orientation, order, circular skewness, circular kurtosis, and the angular distribution describe endpoint axes after the global filament filters. Direction Changes per μm and the orientation maps describe local directions along the full processed skeleton. Filaments can therefore share a common overall orientation while bending locally, or remain individually straight while pointing in different directions.

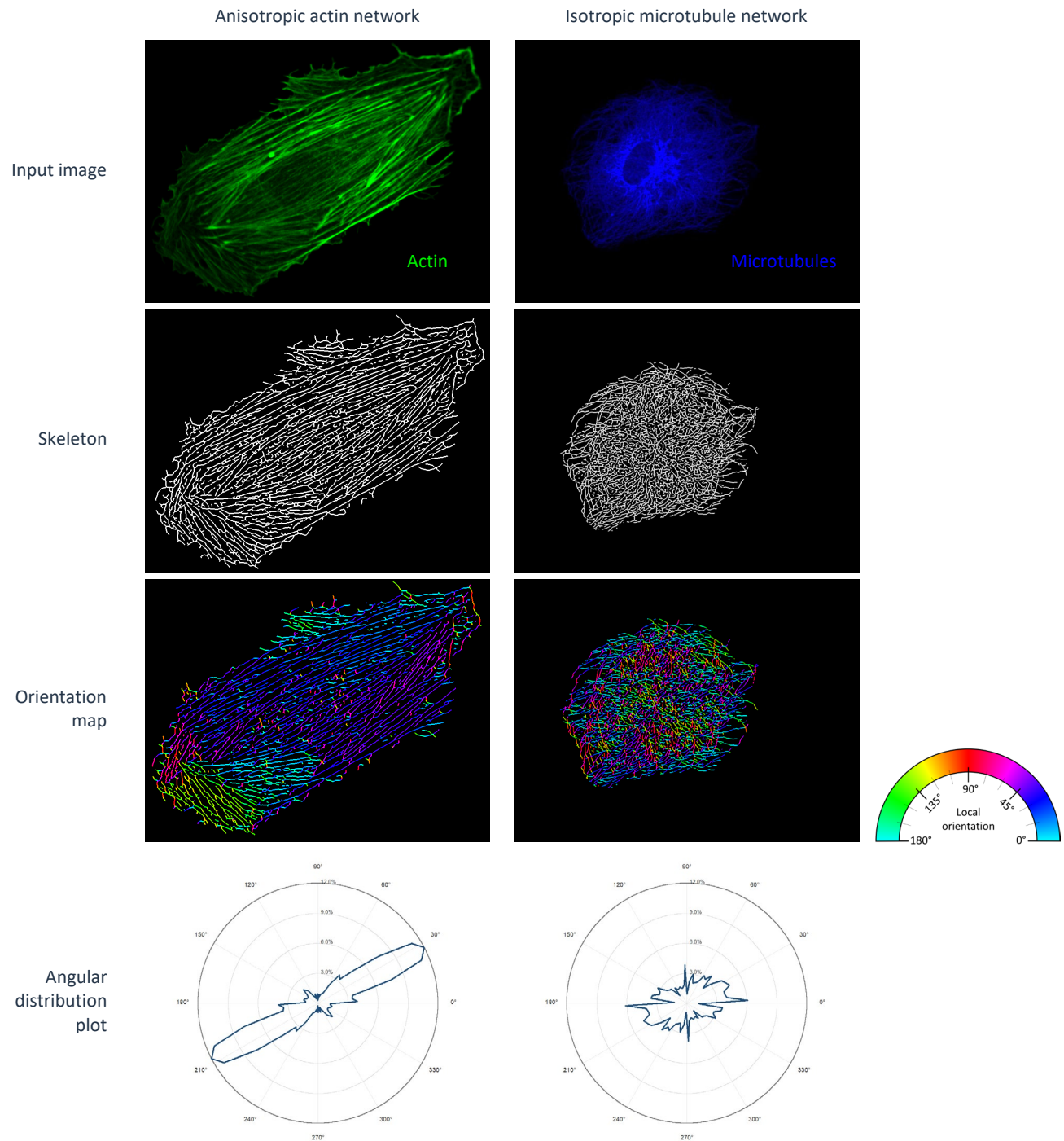


Figure 8. Comparison of anisotropic actin and isotropic microtubule networks.

6 References and resources

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Resources

13. AnalyzeSkeleton plugin documentation. imagej.net/plugins/analyze-skeleton
14. Fiji downloads and system requirements. imagej.net/software/fiji/downloads
15. Fiji Updater documentation. imagej.net/plugins/updater
16. OrientationJ project and installation instructions. bigwww.epfl.ch/demo/orientationj
17. CytoOrg-ImageJ source repository. GitHub. github.com/EP-bioimage-tools/CytoOrg-ImageJ
18. CytoSpectre User Guide. <https://webpages.tuni.fi/cytospectre/>

Images

The fluorescence microscopy images used in this guide were obtained from [idr0050-springer-cytoskeletalsystems/experimentA](https://www.ebi.ac.uk/bioproject/1000000) in the [Image Data Resource \(IDR\)](https://www.ebi.ac.uk/bioproject/1000000). The dataset comprises bright-field and immunofluorescence images of human umbilical vein endothelial cells (HUVECs) showing actin, vimentin, and microtubules. Copyright remains with Springer et al.; the dataset is distributed under the Creative Commons Attribution 4.0 International license ([CC BY 4.0](https://creativecommons.org/licenses/by/4.0/); data DOI: [10.17867/10000119](https://doi.org/10.17867/10000119)). Images used in this guide were processed and annotated to illustrate CytoOrg analyses.

CytoOrg was initially developed using immunofluorescence images of γ -actin in human SK-MEL-31 melanoma cells. These images are not shown in this guide because the work has not yet been published.

7 Troubleshooting and feedback

If a run fails, note the error message and last image named in the Fiji Log, then check the relevant points below.

The script does not start, or a command is not recognized. Open the .groovy file in the Fiji Script Editor and select Groovy. Check that AnalyzeSkeleton, Local Thickness, and Tubeness are available. If OrientationJ Analysis is not recognized, install OrientationJ. Apply Fiji updates, restart, and test one image before running the full dataset.

No valid original image files are found. Select the folder containing the images themselves. CytoOrg reads TIF, TIFF, PNG, JPG, and JPEG files, but does not search subfolders. Names beginning with segcell_ are reserved for masks.

A mask is missing, has the wrong size, or contains no cell ROI. Check the segcell_<complete original filename> naming, including the extension, and place the mask beside its original image. Width and height must match. Use 8-bit binary masks with cells at intensity 255 and background at 0. CytoOrg stops if a mask is missing, dimensions differ, or no cell is detected.

Cells are merged, split, or counted incorrectly. Each cell must be one connected white object, separated from other cells even at corners. Remove debris and unintended fragments. Verify segmentation using the ROI overlay. For single-cell inputs, remove all signal outside the intended cell and check that its outline is continuous.

Folders or output files cannot be written. The input folder must be writable: both results and temporary multi-cell files are created inside it. Check free disk space, close output files open in other applications, and try a local folder. Use a simple output folder name without path separators or invalid characters.

The run stops during thickness analysis. If no values survive the intensity threshold and minimum-thickness cutoff, the current script cannot calculate quartiles and may stop with an array-index error. Check the 8-bit signal, pixel size, and thresholds. If thickness is not required, disable both thickness measurement and automatic sigma estimation.

The skeleton is empty, fragmented, or excessively connected. Compare it with the original image. Check signal quality, pixel size, Tubeness sigma, minimum length, and pruning. High cutoffs or endpoint pruning can remove genuine structures; inappropriate sigma or background can create false connections. Validate settings on representative images.

Some results are NaN, or the angular plot has no distribution line. No filament may pass the filters, or a denominator may be zero. For orientation, check minimum length, maximum tortuosity, and side-branch exclusion. Rebranching rate is undefined when no side branches are detected. Treat undefined values as missing data, not zero.

An expected map or table is missing. Check its analysis and export options. By default, enhanced branching and orientation maps are saved without the standard maps. With at least 10 analyzed cells or images, outputs are placed in category subfolders; Results.csv remains in the main results folder. A missing final table may indicate interruption.

Interrupted runs and recovery

Preserve completed work before restarting. Images and plots are saved during processing, but Results.csv is written only after the image loop finishes. Save the open Cytoskeletal Organization - Results table under a new filename before closing Fiji or starting another run. A crash may lose unsaved rows.

Restart only the unfinished part. CytoOrg does not automatically resume an interrupted analysis or append new rows to an earlier results table. Identify completed cells from the saved rows and outputs. Prepare a separate input folder containing the remaining images and use a new output folder name. For partially processed multi-cell images, work on copies of the original images and masks, retaining only the unprocessed cells. Removing cells from a mask will change cell numbering; use the new ROI overlays to match results to the correct cells.

Remove temporary files only after the run has stopped. A `.cytoorg_temp_<timestamp>` folder may remain. Check the outputs and preserve useful ROI overlays before deleting it; never remove it during an active run. If necessary, save required work and restart Fiji. Avoid closing working images, result tables, or the ROI Manager during analysis.

Questions and feedback

Questions, bug reports, feature requests, and suggestions for improving CytoOrg are welcome. Please contact Edoardo Perina at e.perina@studenti.unibs.it

8 License agreement

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